

EXP 8:-Demonstrate Classification, Clustering, Association using weka

Aim:

Perform data Pre-processing task and demonstrate Classification, Clustering, Association algorithm on data sets using data mining tool WEKA

Introduction:

WEKA (Waikato Environment for Knowledge Analysis) is a popular open-source software developed at the University of Waikato, New Zealand. It provides a collection of machine learning algorithms for data mining tasks such as data preprocessing, classification, regression, clustering, association rules, and visualization. WEKA supports a user-friendly graphical interface that simplifies applying various algorithms on datasets without writing code.

The tool operates on datasets in the ARFF (Attribute-Relation File Format) or CSV format and allows users to easily load, process, and analyze data. The strength of WEKA lies in its wide range of implemented algorithms and built-in support for cross-validation, visualization, and model evaluation.

In this experiment, we demonstrate:

- Classification: Predicting categorical labels based on input features using algorithms like J48, Naive Bayes, etc.
- Clustering: Grouping similar data points together without pre-defined labels using algorithms like k-Means.
- Association: Discovering interesting relationships or associations among attributes in large datasets using Apriori algorithm.

Procedure

Step 1: Load Dataset

- Open WEKA GUI Chooser.
- Choose "Explorer".
- Click on "Open file" and load a dataset

Step 2: Preprocess Data

- View attribute summary.
- Remove or transform attributes if required.
- Ensure no missing values or irrelevant fields.

Step 3: Apply Classification

- Go to "Classify" tab.
- Choose classifier (e.g., J48 decision tree, Naive Bayes).
- Select class attribute.

- Click "Start" to train and test the model.
- Analyze results: accuracy, confusion matrix, ROC curve, etc.

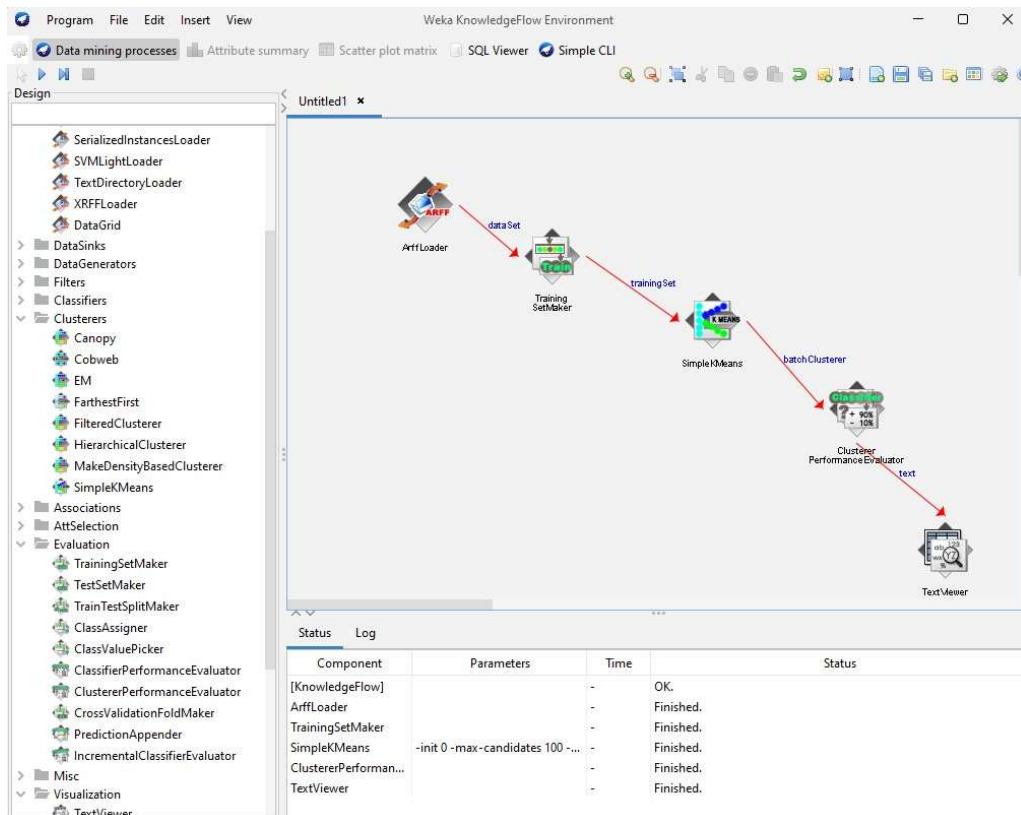
Step 4: Apply Clustering

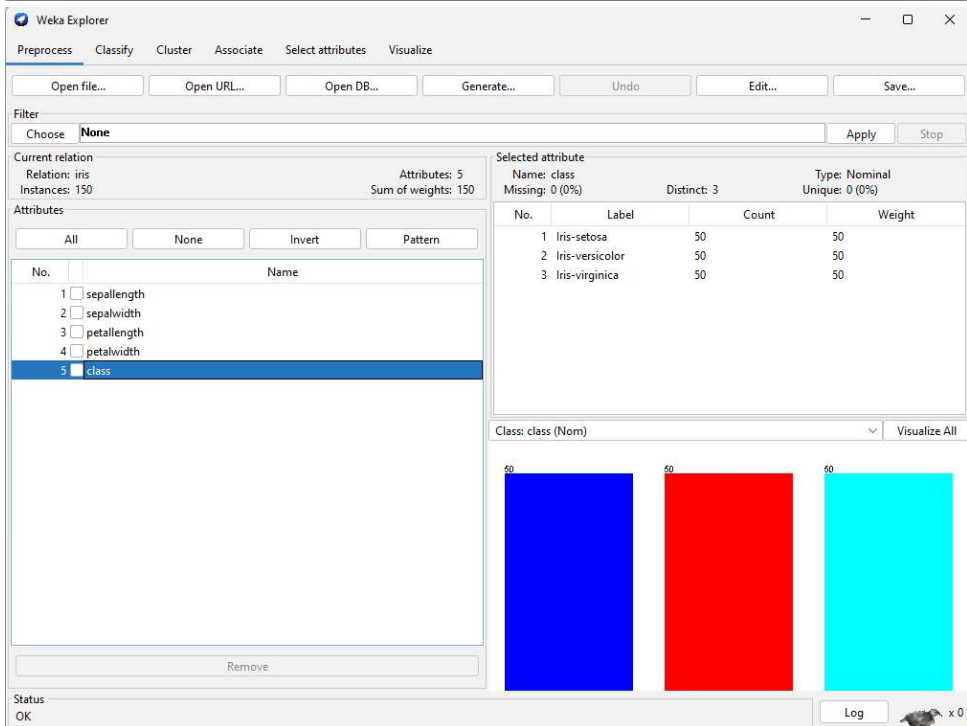
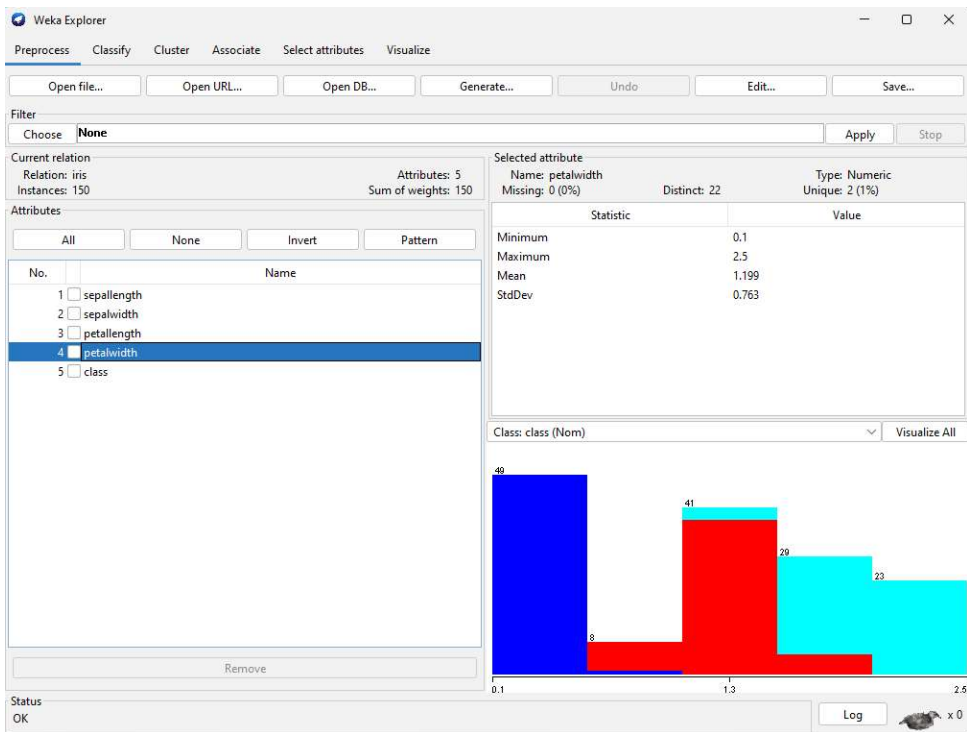
- Go to "Cluster" tab.
- Choose clustering algorithm (e.g., SimpleKMeans).
- Configure number of clusters.
- Click "Start" and view cluster assignments.

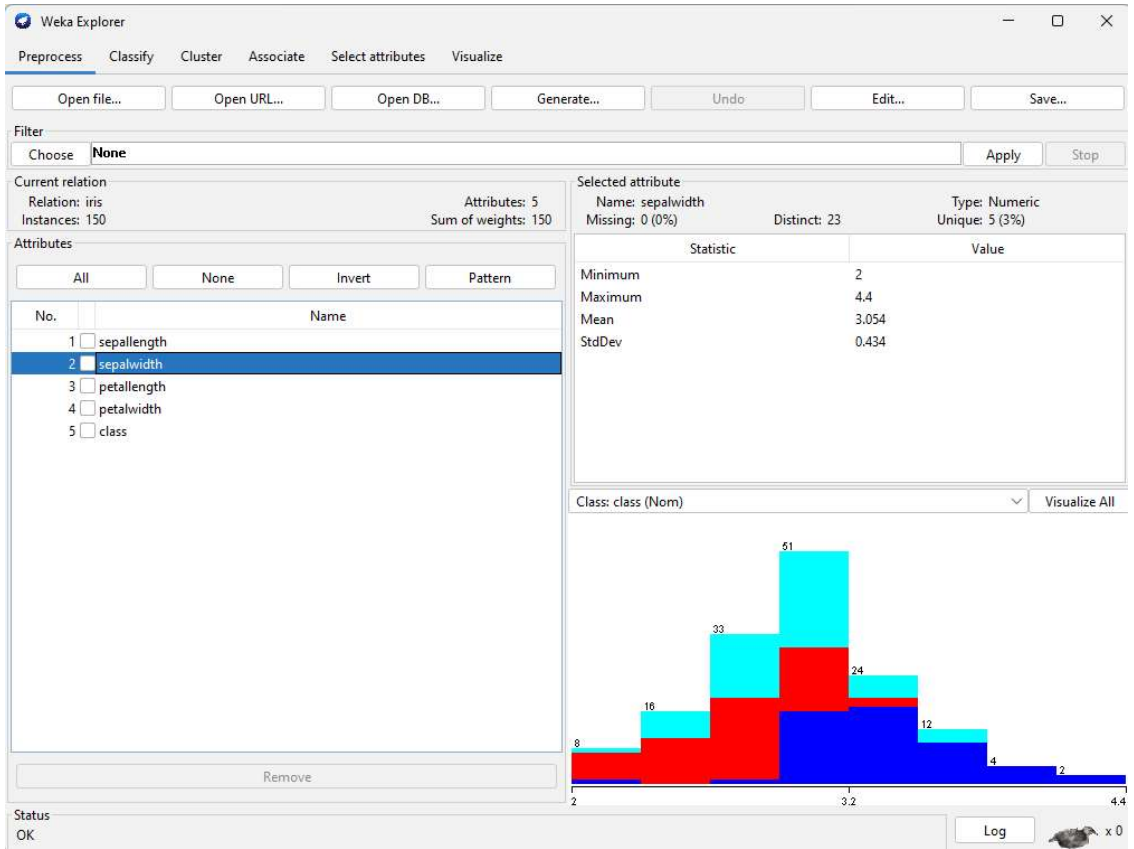
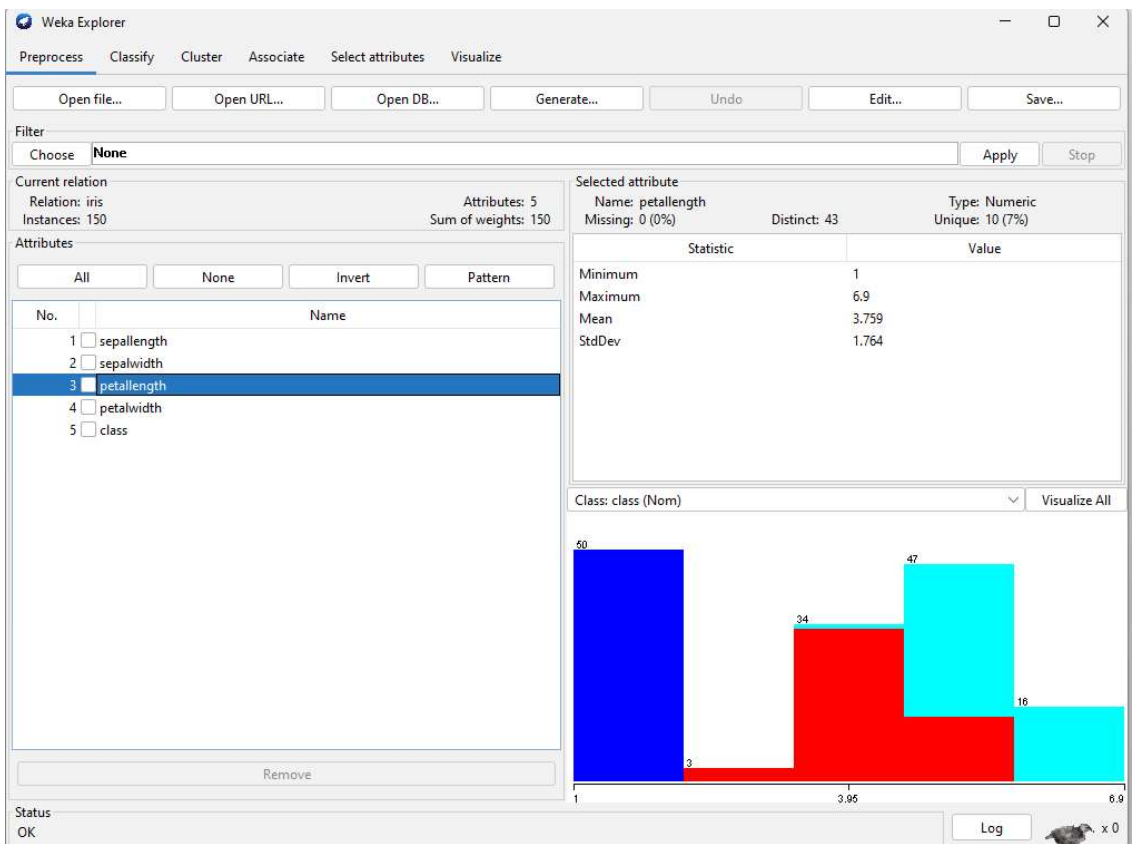
Step 5: Apply Association

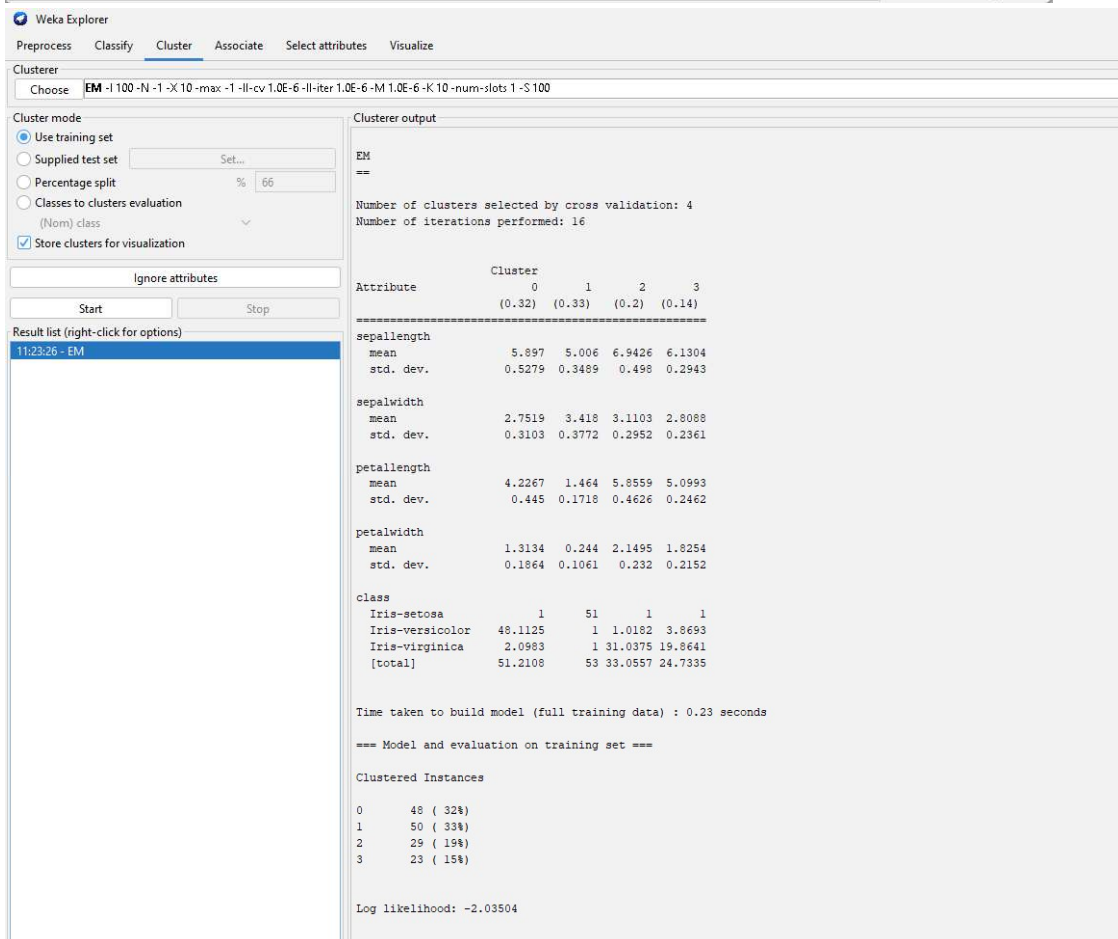
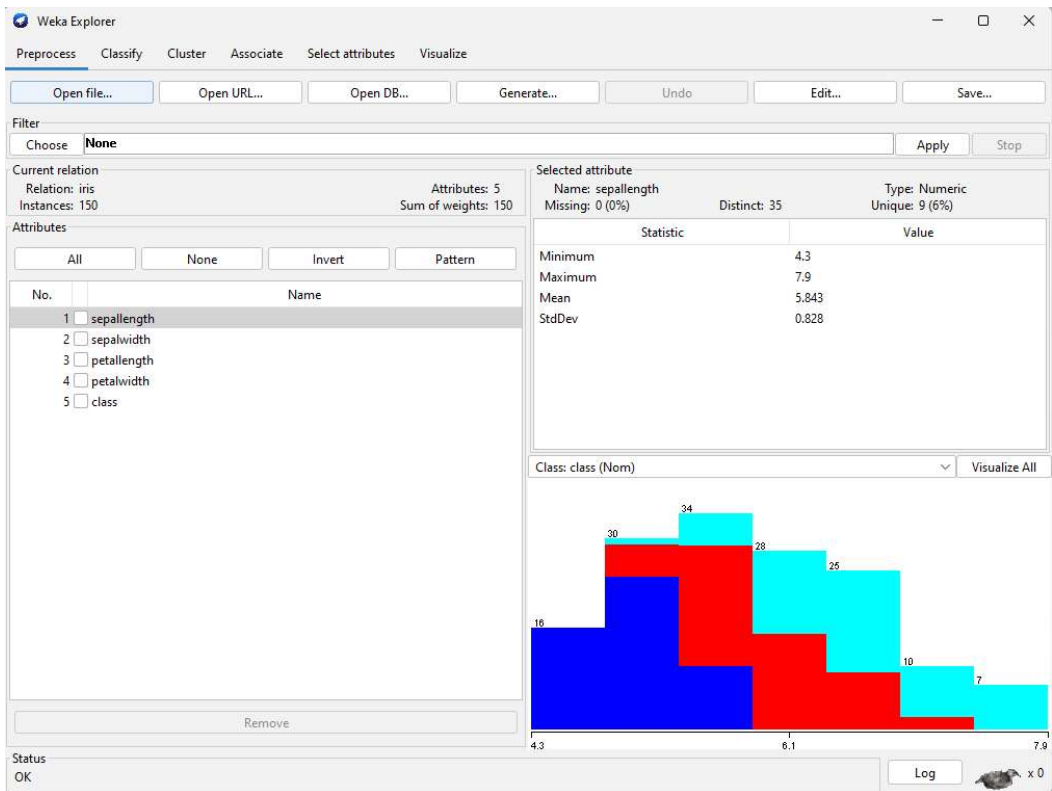
- Go to "Associate" tab.
- Choose algorithm (e.g., Apriori).
- Set minimum support and confidence thresholds.
- Click "Start" and view generated rules.

Output









Weka Explorer

Preprocess **Cluster** Associate Select attributes Visualize

Clusterer: Choose **EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100**

Cluster mode:
☒ Use training set
☐ Supplied test set
☐ Percentage split
☐ Classes to clusters evaluation
☒ Store clusters for visualization
 (None) class

Ignore attributes

Start Stop

Result list (right-click for options)

112326-EM

Clusterer output

==== Run information ====

Scheme: weka.clusterers.EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100
 Relation: iris
 Instances: 150
 Attributes: 5

Test mode: evaluate on training data

==== Clustering model (full training set) ====

EM

Number of clusters selected by cross validation: 4
 Number of iterations performed: 16

Attribute	Cluster			
	0	1	2	3
	(0.32)	(0.33)	(0.2)	(0.14)
sepal.length				
mean	5.897	5.006	6.9426	6.1304
std. dev.	0.5279	0.3489	0.498	0.2943
sepal.width				
mean	2.7519	3.418	3.1103	2.8088
std. dev.	0.3103	0.3772	0.2952	0.2361
petal.length				
mean	4.2267	1.464	5.8559	5.0993
std. dev.	0.445	0.1718	0.4626	0.2462
petal.width				
mean	1.3134	0.244	2.1495	1.8254
std. dev.	0.1864	0.1061	0.232	0.2152
class				
Iris-setosa	1	51	1	1
Iris-versicolour	48.1125	1	1.0182	3.8693
Iris-virginica	2.0983	1	31.0375	19.8641
[total]	51.2108	53	33.0557	24.7335

Conclusion

In this experiment, we explored how to use the WEKA tool to apply various data mining techniques, including classification, clustering, and association rule mining. We learned how to load and preprocess data, apply different algorithms, and interpret the results effectively. WEKA provides a simple yet powerful interface for applying machine learning models and is a great educational tool for understanding core concepts in data mining.